

STORIES, STATISTICS, AND MEMORY *

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For many decisions, we encounter relevant information over the course of days, months, or years. We consume such information in various forms, including stories (qualitative content about individual instances) and statistics (quantitative data about collections of observations). This article proposes that information type—story versus statistic—shapes selective memory. In controlled experiments, we document a pronounced story-statistic gap in memory: the average impact of statistics on beliefs fades by 73% over the course of a day, but the impact of a story fades by only 32%. Guided by a model of selective memory, we disentangle different mechanisms and document that similarity relationships drive this gap. Recall of a story increases when its qualitative content is more similar to a memory prompt. Irrelevant information in memory that is similar to the prompt, on the other hand, competes for retrieval with relevant information, impeding successful recall. *JEL codes:* D01, D83.

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I. INTRODUCTION

People accumulate a wealth of information over the course of time, ranging from stories (qualitative descriptions of individual instances) to statistics (collections of information shown in numbers).¹ However, when making decisions, people may remember only a subset of all relevant information. In this article, we examine the nature of such selective memory and its influence on the evolution of beliefs. We hypothesize that people are more likely to successfully remember stories than statistics, meaning that their beliefs are more persistently influenced by information they receive in the form of stories.

To study the temporal evolution of beliefs in response to different types of information, we run a series of tightly controlled, preregistered experiments. In our baseline experiment, a hypothetical product or venue has received a number of reviews, each either positive or negative. We randomize what people learn about the number of positive reviews. In the *Statistic* condition, respondents receive quantitative information about the number of positive reviews in a randomly drawn subsample of reviews. In the *Story* condition, respondents obtain information about a randomly drawn single review alongside qualitative content describing the experience underlying the review. In a third condition, respondents receive no additional information. The main outcomes of interest are participants' incentivized beliefs about whether another randomly drawn review is positive. Each participant completes three independent scenarios about different products and venues and is assigned to each information condition once. To examine the role of memory, we elicit beliefs from participants twice: immediately after receiving the information (the *Immediate* condition) and again following a one-day delay (the *Delay* condition). The temporal structure is crucial to our design because there are numerous differences between stories and statistics that could result in different beliefs; however, any such differences not arising from memory constraints are accounted for by the immediate belief update. Therefore, since no new information is received in the interim, any change in stated beliefs over time must, by design, be due to memory.

1. This is the Oxford dictionary's definition of a statistic.

We document that in line with imperfect memory, average beliefs partially revert to the prior as time passes. This temporal decay in beliefs emerges for both types of information. Our main finding is a pronounced story-statistic gap in the evolution of beliefs: the effect of stories on beliefs decays less strongly than the effect of statistics. Pooling all statistics and stories presented in our baseline study, we find that on average, the magnitude of belief reversion from the immediate update toward the prior is more than twice as large for statistics (73%) as for stories (32%). In fact, we find that the average belief impact (the difference between a stated immediate or delayed belief and the induced prior) is larger for statistics than stories in Immediate, but smaller in Delay. This means that the relative magnitudes of belief impact flip over time: statistics are, on average, interpreted as more informative in the moment and shift beliefs more strongly, but as time passes this effect reverses and the effect of stories on beliefs is larger. To provide direct evidence on the role of selective memory, we use a free-recall task in the follow-up survey. We find that participants are approximately twice as accurate at recalling the correct type and direction of the information for the scenario in which they received a story than for the one in which they received a statistic.

Additional experiments demonstrate the robustness of our findings in a setting where respondents receive stories with qualitative content that is known to be uninformative. This allows us to compute the Bayesian benchmark based on the quantitative information contained in stories and statistics, respectively. Our experiments replicate our previous findings and establish that immediate belief updates are fairly close to Bayesian benchmarks for both types of information. Moreover, we conduct an experiment that shows that adding uninformative qualitative content to statistical information also significantly decreases belief decay and increases accurate recall. This suggests that the qualitative features of stories are what drive the story-statistic gap in memory.

To guide our investigation of underlying memory mechanisms, we propose a simple model of selective memory that adapts [Bordalo et al. \(2023b; forthcoming\)](#) to accommodate stories and statistics. Our framework follows canonical models of memory, where experiences are stored as memory traces in episodic memory. A central feature of episodic memory is its cue-dependent nature: the recall of memory traces depends on their relationship to

a so-called memory cue. In our model, the memory cue originates from an experimental prompt. In our experiments, this prompt consists of the name of the product and a belief question. The prompt evokes *semantic associations* that are related in meaning. The prompt and these associations constitute the memory cue. To illustrate this, consider a scenario from our experiment about a restaurant. The prompt in the “restaurant” scenario is connected in meaning to concepts such as dinner, food, and drinks, all of which are part of the cue. Given this memory cue, recall is stochastic: the decision maker either recalls the relevant target memory or they accidentally recall an irrelevant, nontarget memory. Whether a relevant or an irrelevant memory is retrieved is governed by two similarity relationships. The more similar a target memory trace is to the memory cue, the higher the chances of retrieving it. The more similar nontarget memory traces are to the memory cue, the higher the chances of failing to retrieve the target and accidentally retrieving an irrelevant memory. This latter mechanism is referred to as interference.

Our simple framework makes three testable predictions about the likelihood of successful recall and the temporal decay of the initial belief movement induced by a piece of information. First, it predicts a story-statistic gap: stories are more likely to be successfully retrieved than statistics and are subject to less belief decay. This is because—unlike statistics—stories include qualitative content that is related in meaning and thus similar to the prompt. Second, our model predicts that increasing cue-target similarity by adding elements to an experimental prompt that are related in meaning to a given piece of information increases recall likelihood and decreases belief decay when presented with that prompt. Third, the model implies that increasing the similarity between a cue and nontarget information compromises the likelihood of successful recall and increases belief decay. Specifically, a nontarget story that is related in meaning to the prompt makes it more likely that it will accidentally be retrieved, thereby increasing interference.

We develop experimental tests of the model’s predictions. First, we consider the core mechanism from our model driving the story-statistic gap in memory: similarity between the target memory and the cue. Intuitively, a story might be particularly likely to be retrieved when its content is semantically related to the cue. On the other hand, for statistics, there is generally no semantic relationship to cues, since numbers are not typically

related in meaning to specific concepts. To investigate the relevance of cue-target similarity, our mechanism experiment varies the similarity between the prompt and the story content. We document that the more similar the prompt is to the story content, the higher the accuracy of recall. Second, to test the model's prediction that similarity of irrelevant information to the cue shapes recall, we conduct an experiment that varies the similarity of the prompt to nontarget stories. Consistent with the model's prediction, higher similarity of nontarget stories to the prompt decreases recall and the persistence of belief impact of the target information. These results imply that in environments where many similar but conflicting stories circulate, stories lose their edge over statistics as a communication device. Taken together, our mechanism experiments highlight the importance of both cue-target similarity and interference. Our findings suggest that to persistently shape beliefs, effective communication should focus on qualitative features that are strongly semantically associated with the prompt the audience will likely encounter in the future.

We conclude by examining the relative importance of two potential margins of selective memory: first, people may fail to retrieve any relevant memories for a given scenario; second, people may successfully recall relevant memory traces but only partially recover the original information content. For example, people may remember that the majority of reviews were positive, but not what exact fraction were positive. To connect the magnitude of belief impact to different recall patterns, we jointly examine our recall and belief data. We document that conditional on correct recall, there remains virtually no story-statistic gap. These analyses provide an empirical foundation for modeling selective recall as primarily arising from retrieval failures rather than from distortions in retrieved content.

Our work relates to a nascent literature on stories and narratives in economics (Shiller 2017, 2020; Michalopoulos and Xue 2021; Morag and Loewenstein 2021; Andre et al. 2022b, 2022a; Kendall and Charles 2022; Barron and Fries 2023; Graeber, Noy, and Roth 2024; Graeber, Roth, and Schesch 2024). This literature mostly focuses on the persuasive effects of narratives in moral or political domains (Bénabou, Falk, and Tirole 2018; Eliaz and Spiegler 2020; Bursztyn et al. 2023b, 2023a; Alesina, Miano, and Stantcheva 2023). A related literature in psychology and management studies the power of stories in influencing people (Bruner 1987; Fryer 2003; McAdams 2011; Monarth 2014). We add to the

literature by (i) comparing the effect of stories to that of statistics over time, and (ii) providing systematic, theory-guided evidence on mechanisms with a focus on the role of qualitative information. Our evidence highlights one mechanism by which narratives are effective: they promote recall and thus more easily come to mind at the time of decision-making.

Our work also ties into a growing literature on the role of experiences, attention, and memory in economics (Gennaioli and Shleifer 2010; Malmendier and Nagel 2011, 2016; Bordalo, Gennaioli, and Shleifer 2020; Bordalo et al. 2020, 2021, 2023a; Link et al. 2023). The model heavily builds on Bordalo et al. (2023b, forthcoming) who provide theoretical frameworks in which agents form beliefs by retrieving experiences from memory based on similarity and interference. Enke, Schwerter, and Zimmermann (2024) empirically study the role of associative memory for belief formation and show that it can give rise to overreaction to news. In contrast to our focus on the decay of belief impact over time, Enke, Schwerter, and Zimmermann (2024) examine the extent to which the strength of immediate updating in response to new signals is influenced by the history of previous signals. Afrouzi et al. (2023) experimentally study the role of working memory in forecasting experiments. A series of recent papers provide evidence on the role of associative recall in field settings, for example, in finance (Charles 2022; Jiang et al. 2022; Kwon and Tang 2023) and the labor market (Conlon and Patel 2022). Our article strongly suggests that people do not continuously update their beliefs every time they receive a piece of information; instead, they partly construct them on the fly, consistent with a growing body of evidence on cue-dependent belief formation (Bordalo et al. 2021; Andre et al. 2022b; Enke, Schwerter, and Zimmermann 2024; Bordalo et al. forthcoming). This study differs from this previous literature in its focus on how different types of information—statistics versus stories—shape beliefs over time.

More broadly, our work builds on extensive psychology literature on memory (Schacter 2008; Kahana 2012; Baddeley, Eysenck, and Anderson 2020). Some previous work in psychology directly relates to the recall of stories, but with a particular focus on the role of scripts (Schank and Abelson 1977; Brewer and Treyens 1981; Mandler 1984; Heath and Heath 2007), emotions (Kensinger and Schacter 2008), and mental imagery (Shepard 1967; Standing 1973; Shepard and Cooper 1986). Bower and Clark (1969) document that students' ability to remember a list

of words strongly increases when instructed to create a coherent narrative that contains all of the words.² These papers differ from ours in a number of ways. First, they focus on studying the recall of word lists but do not measure beliefs or track their evolution over time. Second, they do not compare the dynamics of belief formation based on statistics versus stories. Finally, these experiments do not aim to tightly identify underlying cognitive mechanisms, such as the role of cue-target similarity or interference, which are crucial ingredients for models of cue-dependent memory (Bordalo et al. 2023b, forthcoming).

This article proceeds as follows: in Section II, we outline a simple model of selective memory that formalizes mechanisms driving differences in the recall of stories versus statistics. Section III presents experiments that demonstrate the existence and robustness of a story-statistic gap in memory. Section IV describes our evidence on mechanisms. In Section V, we provide a decomposition of the story-statistic gap, and Section VI discusses the implications of our findings.

II. A MODEL OF SELECTIVE MEMORY

II.A. Setup

We outline a model of memory that adapts Bordalo et al. (2023b, forthcoming) to formalize cue-dependent recall and belief formation for stories and statistics. The model setup mirrors our experimental paradigm. Consider a decision maker (DM) who learns about the reviews a specific product or venue has received. There is a population of N reviews, each of which is either positive or negative. The DM enters with a uniform belief over the number of positive reviews among N (ignorance belief). There are two periods. In the first period, the DM may receive additional information about the reviews of the product, either in the form of a story or a statistic. We define a statistic as a randomly drawn subset n of N that includes k positive and $n - k$ negative reviews. For our baseline setup, we define a story as a statistic of $n = 1$ complemented with additional nonquantitative content, akin to

2. This relates to techniques for memory enhancement, which use visualizations of familiar spatial environments to improve the recall of information, commonly referred to as memory palaces or method of loci (Foer 2012).

an anecdote about a single review.³ In the second period, the DM receives no additional information. In both periods, the DM states a belief that a randomly drawn review from N is positive.

Over the course of the experiment, the DM faces three scenarios, each one about a different product or venue. Across these scenarios, the DM receives one story, one statistic, and once no additional information.

II.B. Similarity and Recall

In canonical models of memory, personal experiences are stored in episodic memory (Kahana 2012). We assume that the DM's episodic memories are organized in a memory database M . Each element of M is a memory trace m that encodes one experience. A trace is a vector of $F \geq 1$ features with values in $V_1 \times \cdots \times V_F$. Some sets of possible values V_f contain the null value $\emptyset$, indicating the absence of a feature. Recall is cue-dependent, which means that recall is initiated by an external trigger that may be a situation, question, or specific event. We represent a cue in the same way as memory traces, as a vector of length F with entries in $V_1 \times V_2 \times \cdots \times V_F$. Given any database of traces and any cue, recall is stochastic and governed by the similarity relationships between the cue and the memory traces. We define a similarity measure over any two traces or cue vectors, x_1 and x_2 ,

$$S(x_1, x_2) : \prod_{i=1}^F V_i \times \prod_{i=1}^F V_i \rightarrow [0, 1],$$

and require that it is symmetric, increasing in the number of features that share the same value, equals one if and only if $x_1 = x_2$ and equals zero if and only if no feature is shared. The probability of recalling a specific target trace m^* when cued with c is given by

$$(1) \quad r(m^*, c) := \frac{S(m^*, c)}{\sum_{m \in M} S(m, c)}.$$

The probability of recall is jointly governed by the cue-target similarity in the numerator, that is, the similarity between the cue

3. However, note that the addition of qualitative content is, in principle, independent of the sample size of the corresponding statistic, which we explore experimentally in Section III.E.

and the target trace m^* , and interference in the denominator, that is, the similarity between the cue and all other memories. The likelihood of successfully recalling the target trace increases in the cue-target similarity. On the other hand, higher interference yields a higher likelihood of accidentally retrieving an irrelevant memory trace.

II.C. Memory Traces and Cues

1. *Memory Traces.* In the baseline experiment, there are three scenarios: a bicycle, a restaurant, and a video game, creating three corresponding memory traces. We assume the following vector structure for memory traces: the first dimension identifies the scenario, that is, $V_1 = \{\text{bicycle, restaurant, video game}\}$ in the baseline experiment. The second entry encodes the number of reviews n that the DM learns about, $V_2 = \{0, \dots, N\}$, with zero implying that no additional information was received. The third entry carries the number k of positive reviews among the n reviews provided, that is, $V_3 = \{0, \dots, N\}$, with $k \leq n$. All additional entries of a trace represent the nonquantitative content provided through a story. These dimensions encode everything that was mentioned in the qualitative content of the scenario. We refer to this set of features by V^{qual} and do not constrain its structure further. V^{qual} is large enough to encode any possible story across all scenarios, and each feature takes the null value or 1, indicating the absence or existence of that feature.⁴ For any given trace m , we refer to a realization of the qualitative features as $V^{\text{qual}}(m)$.

Note that we focus on how memory traces encode the content that was explicitly provided in a scenario. In addition, standard memory models assume that episodic memories also encode the context in which an experience was made, such as the time of day or weather. Such context could be readily accommodated by $V^{\text{qual}}(m)$, but because most context is plausibly shared across the experiences associated with different scenarios in the experiment, it is largely irrelevant for our purposes besides introducing a base-level similarity between all traces. This is why our exposition of the trace structure focuses on content rather than context.⁵

4. For modeling reasons, all memory vectors share the same dimensionality, that is, all have $V^{\text{qual}} + 3$ entries.

5. It is possible that encoding depends on the informativeness of the information. We abstract from such differences for the sake of simplicity.

Memory traces associated with stories and statistics both contain entries in the first three dimensions. The key difference is that stories provide additional nonquantitative content, captured by $V^{\text{qual}}(m)$. A memory trace for a scenario has at least one feature present in V^{qual} if a story was received, but none if a statistic or no additional information was received in the baseline experiment.

In what follows, we denote the treatment type of a memory trace (statistic, story, or dummy) in superscript and the product type in subscript. Given the above assumptions, a statistic conveying that three out of seven reviews for the scenario bicycle were negative forms the following trace in the memory database:

$$m_{\text{bicycle}}^{\text{statistic}} = (\text{bicycle}, 7, 4, V^{\text{qual}}(m_{\text{bicycle}}^{\text{statistic}}))$$

$$\text{with } V^{\text{qual}}(m_{\text{bicycle}}^{\text{statistic}}) = (\emptyset, \emptyset, \dots).$$

A story about a negative review at a restaurant would enter the database as

$$m_{\text{restaurant}}^{\text{story}} = (\text{restaurant}, 1, 0, V^{\text{qual}}(m_{\text{restaurant}}^{\text{story}}))$$

$$\text{with } V^{\text{qual}}(m_{\text{restaurant}}^{\text{story}}) \neq (\emptyset, \emptyset, \dots),$$

that is, $V^{\text{qual}}(m_{\text{restaurant}}^{\text{story}})$ does not only contain null entries. In particular, $V^{\text{qual}}(m_{\text{restaurant}}^{\text{story}})$ contains some entries that represent nonquantitative attributes of the story; for example, that the food was stale or the waiter was unfriendly. The trace produced in a scenario about a video game where no additional information was provided would be encoded as

$$m_{\text{video game}}^{\text{dummy}} = (\text{video game}, 0, 0, V^{\text{qual}}(m_{\text{video game}}^{\text{dummy}}))$$

$$\text{with } V^{\text{qual}}(m_{\text{video game}}^{\text{dummy}}) = (\emptyset, \emptyset, \dots).$$

2. *Cues.* Retrieval is triggered by a memory cue, which in our setup originates from a prompt in the experiment. The prompt contains the question about a randomly drawn review of the product and the scenario name. Unlike the prompt, which is an experimentally controlled feature, the cue is a theoretical construct. We formalize the cue as invoking a vector c that contains both the prompt as well as a (potentially large) set of semantic associations with the prompt. These are connections and

relationships that words or concepts have in common with the prompt based on their meanings (McRae and Jones 2013). The intuition is that the prompt triggers connected concepts that automatically come to mind. For instance, when reading the word “restaurant,” natural semantic associations may be “food,” “service,” or “atmosphere.” We abstain from modeling the structure of semantic associations, which is outside the scope of our article. Instead, we only assume that for any scenario s , the experimental prompt evokes a cue-vector c_s that includes nonnull features in V^{qual} .

Moreover, we assume that a story is actually about the scenario at hand; in a natural, relevant story, at least parts of its content are semantically associated with the scenario. The story relates in meaning to the underlying situation. As a result, the memory trace formed by reading a story includes some features that overlap with what the DM semantically associates with the prompt alone. Formally, there is at least one shared feature between $V^{\text{qual}}(m_s^{\text{story}})$ and $V^{\text{qual}}(c_s)$. To illustrate, consider a story in which stale sushi was served at a restaurant. This could be represented in the memory trace by a feature encoding (bad) food. Being prompted with the word “restaurant” automatically triggers thoughts of food, so that the cue and the story trace share the feature “food.”

II.D. Belief Formation

The DM forms a belief about whether a randomly drawn review in a given scenario is positive. We remain agnostic about how exactly beliefs are formed given the information that is presented or recalled and hence formalize a very general updating rule that only assumes time invariance.⁶ All the distortions we study experimentally are captured by the recall process.

Entering with an ignorance belief, the DM (potentially) receives additional information on a scenario in the first period. They form a posterior belief and, at the same time, store a single memory trace m^* that follows the structure outlined above. In the second period, the DM is again prompted to state their belief. Rather than recalling their first-period posterior directly, we propose that the DM again enters this decision with their ignorance belief, and may or may not remember the additional

6. This general updating rule nests Bayesian updating.

information received in the meantime. In terms of timeline, this can be thought of as the DM coming in with a flat belief, accumulating information over time, and then trying to remember this past information in the moment they face a decision. The prompt generates a cue c , which gives the agent a chance $r(m^*, c)$ to recall the target trace m^* and with it all the relevant additional information from the scenario. If any other trace than the target trace is retrieved, the agent notices their mistake and discards it. Successful recall leads to a posterior belief in the second period that is identical to the first-period belief, whereas failed recall means the agent reverts to their ignorance belief.

1. *Notation.* For a given scenario, the DM's belief in period t is π_t and the stated belief is $\hat{\pi}_t$. The ignorance belief is denoted by π_0 . We define a general updating rule Φ as a function taking the ignorance belief π_0 and an information set I and yielding a posterior belief π_1 .

2. *Stated Belief Absent Additional Information.* We assume that if the DM does not have access to additional information, the posterior equals the ignorance belief, that is, $I = \emptyset \Rightarrow \pi_0 = \pi_1$. The experiment implements a scoring rule under which reporting the mean maximizes payoffs:

$$\hat{\pi}^{noinfo} = \mathbb{E}[\pi_0] = \frac{1}{2}.$$

3. *Stated Beliefs with Additional Information.* The DM forms a belief update using the updating rule Φ , ignorance belief π_0 , and the additional information I to arrive at belief π^{info} . Given the incentive structure, the DM then states their expected value $\hat{\pi}^{info} = \mathbb{E}[\pi^{info}]$.

4. *Recall and Belief Decay.* We formalize belief decay as the absolute value of the difference between the beliefs formed in the second and first period. Note that if recall is successful, beliefs are stable so that $\hat{\pi}_2 = \hat{\pi}^{info} = \hat{\pi}_1$, and if recall fails, then $\hat{\pi}_2 = \hat{\pi}^{noinfo}$. The expected second-period belief conditional on period 1 is $\mathbb{E}[\hat{\pi}_2 | \hat{\pi}_1] = r(m^*, c)\hat{\pi}_1 + (1 - r(m^*, c))\frac{1}{2}$. Belief decay is governed by the probability of recall scaled by the distance of the

first-period belief to the ignorance belief:

$$(2) \quad \mathbb{E}[|\hat{\pi}_2 - \hat{\pi}_1| \mid \hat{\pi}_1] = (1 - r(m^*, c)) \cdot \left| \frac{1}{2} - \hat{\pi}_1 \right|.$$

5. *Discussion.* In our model, beliefs are proportional to recall. Indeed, there are only two possible beliefs people hold in our model, even after a delay: the posterior belief formed in period 1 and the ignorance belief. We consider this feature a strong simplification for the purpose of demonstrating the key implications of our model. We acknowledge that belief formation data in practice typically exhibit noise and are unlikely to be fully explained by our model that predicts a bimodal distribution. Our analyses in [Section V](#), however, demonstrate that a large majority of often up to 80% of our observations are indeed captured by either no belief decay or full reversion to the ignorance belief, justifying our model simplification. A more complete model would incorporate noise in the belief formation or retrieval process, along the lines of [Enke and Graeber \(2023\)](#). Notice that under our model, which only accommodates two possible updates, we cannot identify the degree of belief decay at the individual level but only across subjects. This does not constrain our empirical analyses much, nor does partial belief decay at the individual level play a large role in our data, as discussed in [Section V](#).

II.E. Predictions

Our simple framework makes three main predictions that guide the empirical analysis. The first one establishes the existence of a story-statistic gap. All derivations and additional predictions are relegated to [Online Appendix G](#).

PREDICTION 1. (*Story-Statistic Gap*) *The likelihood of successful recall is higher for stories than for statistics. Conditional on first-period beliefs, belief decay for stories is lower than for statistics.*

Intuitively, recall of stories is more likely than that of statistics because the additional, nonquantitative content is semantically associated with the scenario, and thus stories exhibit a higher cue-target similarity than statistics. A higher likelihood of successful recall induces less belief decay in expectation.

The next two predictions concern the building blocks of the recall mechanism in [equation \(1\)](#): cue-target similarity and

interference. As to the former, we obtain the following implication on the impact of changing the prompt in a way that increases the number of shared features.

PREDICTION 2. *(Cue-Target Similarity) Changing the prompt to invoke semantic associations that have a larger overlap with the target memory trace raises cue-target similarity. This increases the likelihood of successful recall and decreases belief decay.*

Interference is governed by the similarity of nontarget memories m to the cue c , $S(m, c)$. The higher the similarity between the qualitative content of nontarget traces and a given target cue, the more pronounced are forgetting of the target trace and belief decay. To illustrate, assume that there are two scenarios, one about a food truck and one about an amusement park. Consider a story provided in the amusement park scenario that is either about the rides or about the food consumed in the park. The latter story is naturally more closely related in meaning to the other scenario (food truck) than the former story. The similarity between the given scenario (food truck) and nontarget traces (food in the amusement park) induces a higher probability of accidentally retrieving the memory created for the amusement park in recall about the food truck. In terms of the model, this is reflected in an overlap between $V^{\text{qual}}(m_p^{\text{story}})$ and $V^{\text{qual}}(c_q)$.

PREDICTION 3. *(Interference) All else equal, increasing the similarity between a story in scenario p and a cue for another scenario q decreases the likelihood of successful recall and increases belief decay in q .*

III. THE STORY-STATISTIC GAP IN MEMORY

III.A. Baseline Design

Our baseline design is guided by the following objectives: (i) panel data on beliefs that allow us to study the evolution of beliefs over time without new information arriving in the meantime; (ii) a measure of immediate updating that captures any differences in the effects of stories and statistics that are not memory-related; (iii) a setting in which information both in the form of statistics and stories is common; and (iv) an incentive-compatible belief elicitation. [Online Appendix](#) Table A.7 provides an overview of all experiments.

1. *Task Structure and Timing.* There are three different hypothetical scenarios, each one about some product or venue.⁷ Any given product or venue has received an overall number of reviews, with each review being either positive or negative. In the model in [Section II](#), we assume that the decision maker enters with an ignorance belief. To fix this ignorance belief in our experiments, we truthfully inform respondents that the actual number of positive reviews would be randomly drawn from a uniform distribution, independently for each scenario, inducing a uniform prior. For each scenario, participants then receive either a piece of additional information or no additional information and are subsequently asked to state their guess.⁸ Our main outcome of interest are respondents' incentivized beliefs about the likelihood that a randomly selected review is positive.⁹ To examine the role of memory, we elicit beliefs twice: once immediately on receiving the information (condition Immediate) and once one day later (condition Delay).

2. *Stories versus Statistics.* We vary the type of additional information participants are exposed to within subject and across scenarios. For each scenario, participants receive either statistical information (condition Statistic), anecdotal information (condition Story), or no further information. Randomization is blocked such that across scenarios, each person receives one story, one statistic, and once no additional information. Moreover, the order

7. We chose hypothetical scenarios to preclude that relevant additional information can be gathered outside of the experiment.

8. We included the no information treatment because it adds a natural additional source of uncertainty, namely, uncertainty about whether the respondent actually received relevant information in a scenario. Moreover, the treatment allows us to verify that our respondents understand the setting by checking whether, absent any additional information, they state a belief of 50%.

9. The belief elicitation is incentivized using a binarized scoring rule ([Hossain and Okui 2013](#)) with a prize of \$30. The precise payment formula is: Probability of winning \$30 (in percent) = $\frac{100-1}{100}$ (estimate (in percent) - Truth)², where truth = 100 if the randomly selected review is positive, and 0 if not. The binarized scoring rule is incentive compatible, even in the presence of risk aversion. [Danz, Vesterlund, and Wilson \(2022\)](#) document that empirically, the binarized scoring rule can lead to systematic bias in reported beliefs. Notice that even if such bias were present in our experiment, it would not compromise our identification, which relies on the comparison of beliefs between Immediate and Delay for stories and statistics. Moreover, all of our findings are supported by evidence on recall, which is immune to the concern about scoring rules.

of scenarios is randomized and each individual receives one positive signal and one negative signal.¹⁰

We conceptualize statistics as quantitative information about many reviews. In contrast, we define stories as quantitative information about a single review coupled with qualitative content. Thus, stories and statistics differ along two margins in our baseline setup: first, statistics describe multiple data points, while stories are about only one data point. The second difference is the presence of qualitative content.¹¹

Our design closely adheres to this basic taxonomy. Statistical information is communicated as the number of positive reviews for a randomly selected subsample of the population. The fraction of positive reviews is randomly determined, creating variation in the extremity of statistics. Below is an example of how statistical information is communicated:

13 of the reviews were randomly selected. 4 of the 13 selected reviews are positive, the others are negative.

A story provides information about whether a single randomly selected review is positive or negative, plus a qualitative description of that review. The description consists of six to seven sentences recounting the experience underlying the review. We randomize the valence of the qualitative content described in the review between participants. For our main analysis, we focus on stories in which the valence of the qualitative content matches the overall review rating. Below is a shortened example of a story accompanying a negative review about a restaurant.¹²

One of the reviews was randomly selected. The selected review is negative. It was provided by Justin . . . The raw fish looked stale and the sushi rolls were falling apart on the plate . . . The service was poor: his waiter was rude, not attentive and the food was served after a long wait . . . As they left the restaurant, Justin was very annoyed and thought to himself "I definitely won't be back!"

10. [Online Appendix D](#) provides details on the implementation of the randomization.

11. Note that in principle, qualitative content could also be added to statistical information. While we maintain that a natural distinction between stories and statistics is that they tend to differ in sample size, [Section III.E](#) explores the role of adding qualitative content to statistics of fixed sample sizes larger than one.

12. [Online Appendix C.A](#) reproduces all stories from the baseline experiment.

A notable feature of stories is that they are not easily accommodated in a Bayesian belief-updating framework because the informational content of qualitative statements cannot be quantified in a fully objective way.¹³ For instance, in the example above, the qualitative description of the food arguably allows participants to infer that other reviewers may have had similar experiences. Because we cannot determine the normatively optimal Bayesian inference from such qualitative information, we rely on our Immediate belief measurement to capture how informative participants perceive each story to be.¹⁴ Note that this approach is also not reliant on the assumption that people form their beliefs in accordance with Bayes's rule, which may be commonly violated in practice (Enke and Zimmermann 2019; Martínez-Marquina et al. 2019; Enke 2020; Hartzmark, Hirshman, and Imas 2021; Ba, Bohren, and Imas 2023; Graeber 2023).

3. *Prompt.* A critical design feature is the prompt, which in our setup consists of the scenario name and the subsequent belief question. The prompt hence always specifies the product or venue, and then elicits beliefs about whether a randomly drawn review for that product or venue was positive or negative. Specifically, on the screen of the belief elicitation, in condition Immediate and in condition Delay, respondents first learn about the product or venue that we elicit beliefs about. For example, in the case of the restaurant they learn that “a restaurant has received 19 reviews.” To measure beliefs, we use the following instructions:

Out of all the 19 reviews, another review was randomly chosen, where each of the 19 reviews was equally likely to be selected. What do you think is the likelihood (in %) that this review is positive?

4. *Recall Elicitation.* We provide direct evidence on recall of the additional information about product reviews received in the baseline survey.¹⁵ To do so, after displaying the scenario name,

13. In [Section III.D](#) we provide evidence from a setting in which we can cleanly compare belief movements to a Bayesian benchmark.

14. Our approach therefore also accounts for possible differences in the credibility of the information provided in the Statistic versus the Story treatments.

15. We randomized the order of the belief and recall elicitation. In additional studies we replicate our baseline findings with structured incentivized recall tasks instead of the open-ended questions (see [Section III.D](#)).

we ask our respondents the following unincentivized open-ended question:

Please tell us anything you remember about this product scenario. Include as much detail as you can. Most importantly, please describe things in the order they come to mind, i.e., the first thought first, then the next one etc.

5. *Hand-Coding Scheme.* To analyze the unstructured text data, we design and implement a hand-coding scheme (see all details in [Online Appendix E](#)). The scheme records whether respondents mention the direction and type of information they encountered, and whether they correctly remember these characteristics. It also captures additional features, such as (i) whether respondents in the Story condition mention qualitative features, (ii) whether they correctly recall the exact statistical information, and (iii) whether they recall the belief they stated in the baseline survey. To ensure high quality of the hand-coded data, we proceed as follows. First, we instruct three research assistants on the coding scheme and conduct a series of practice rounds with them. Second, each open text response is independently coded by two of the research assistants. Any potential conflicts are resolved by the third research assistant. We find that the interrater reliability is high: for correct recall of type and direction, we find agreement in 94% of the cases.

6. *Incentives.* Participants were informed in advance that the survey consisted of two parts, with one day in between. We also told participants that the information they receive would be relevant for payoffs one day later. Participants were truthfully informed that the computer would randomly select 10% of participants to receive a bonus payment that would be based on their responses.¹⁶ To avoid hedging between similar questions in the two parts, one of the three scenarios and one of the two parts for that scenario (immediate belief, delayed belief) were randomly selected to count for the bonus payment.

7. *Comprehension Checks.* We implemented an attention check as well as extensive control questions to verify participants' understanding of the instructions. Participation in the survey

16. We paid out close to \$15,000 in bonuses across all of our data collections.

required passing an attention check and answering all control questions correctly within the first two trials. These control questions ensure high levels of understanding of the payoff incentives as well as the signals and prior distribution of draws.

III.B. Data

1. *Sample.* We collected data for the baseline experiment on September 8 (baseline) and September 9 (follow-up) 2022. We recruited participants via Prolific, a survey provider commonly used in social science research (Peer et al. 2022). The average duration of the survey was about nine minutes for the baseline, and five minutes for the follow-up. For the baseline, participants received a completion payment of \$1.55 and for the follow-up they received 90 cents.

Fifteen hundred respondents completed wave 1 of our experiment. Out of those, 1,364 met the inclusion criteria and were invited for the follow-up survey.¹⁷ Of those, 1,035 then completed the follow-up survey. After the prespecified sample restrictions,¹⁸ our final sample consists of 933 participants, corresponding to a completion rate of 68%. Given that the key treatment variation is within-person, the attrition rate is not a threat to the internal validity of our findings. For completeness, we report analyses on attrition rates in [Online Appendix Table A.12](#).

2. *Preregistration.* All experiments in this article were conducted online and preregistered on AsPredicted. The preregistrations include the experimental design, hypotheses, analyses, sample sizes, and exclusion criteria. A link to each preregistration is provided in [Online Appendix Table A.7](#). The full set of instructions can be found at https://raw.githubusercontent.com/cproth/papers/master/SSM_instructions.pdf.

17. We exclude observations of 52 participants who were affected by a technical error in the survey code in wave 1. When the drawn statistic corresponded to a share of 100% positive reviews, no numbers were displayed on the screen. Including these additional 52 participants leaves all results virtually unchanged; see [Online Appendix A.H](#). The only other experiment affected by the coding error is Robustness Experiment 4.

18. We prespecified the exclusion of respondents who indicated having written down the information they received and those updating in the wrong direction in response to statistics.

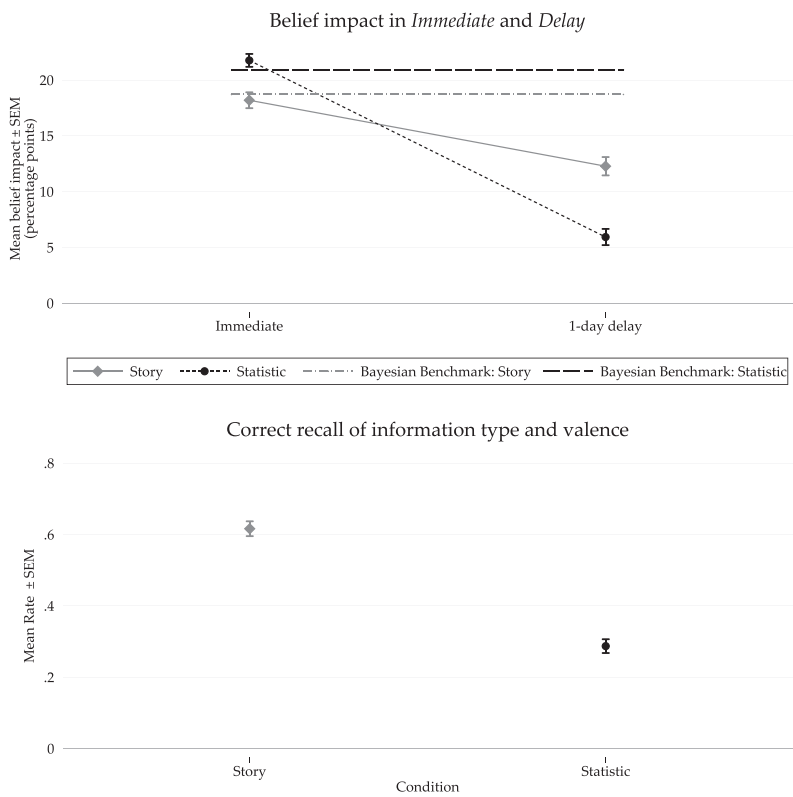


FIGURE I
The Story-Statistic Gap in the Baseline Experiment (933 Respondents)

The top panel displays belief impact in percentage points, separately for conditions Immediate and Delay. Belief impact is the signed distance between a stated belief and the prior (50%). Belief impact is signed in the direction of the rational update. The black markers refer to statistics, while the gray markers refer to stories. The average Bayesian benchmark for statistics is 20.91 percentage points, while the average Bayesian benchmark for stories is 18.71 percentage points. The bottom panel displays the fraction of respondents correctly recalling the type and direction of information they received in the baseline survey, as measured with an open-ended recall question. Whiskers indicate one standard error of the mean.

III.C. Baseline Results

1. *Beliefs.* As preregistered, we start by analyzing stories with qualitative content that is consistent with the overall review rating, which is either positive or negative. The top panel of Figure I and Table I show the average belief impact in Immediate and Delay, pooling the data across scenarios and individuals.

TABLE I
THE STORY-STATISTICS GAP IN MEMORY

Sample	Dependent variable			
	Belief impact			Combined recall
	Immediate (1)	Delay (2)	Pooled (3)	Consistent (4)
Story	-3.57*** (1.24)	6.35*** (1.55)	-3.57*** (1.01)	0.33*** (0.04)
Delay			-15.8*** (0.92)	
Story $\times$ Delay			9.92*** (1.32)	
Control mean	21.76	5.93	21.76	0.29
Observations	1,094	1,094	2,188	1,094
R^2	0.55	0.52	0.43	0.65

Notes. This table uses responses from the Story and Statistic conditions. OLS estimates; standard errors clustered at the participant level are in parentheses. Delay is an indicator taking value one for respondents in the follow-up survey and zero for respondents in the baseline survey. Story takes value one for respondents who received a story for a given product, and zero otherwise. Statistic takes value one for respondents who received a statistic for a given product, and zero otherwise. Columns (1), (2), and (4) include respondents who received consistent stories. Column (3) pools Immediate and Delay. Columns (1) to (3) display results on belief impact. Belief impact is the signed distance between a stated belief and the prior (50%). Belief impact is signed in the direction of the rational update. Column (4) displays the fraction of respondents correctly recalling the type and direction of information they received in the baseline survey. * $p < .10$, ** $p < .05$, *** $p < .01$.

Belief impact is the signed distance between a stated belief and the prior (50%). For ease of exposition, we reverse code the belief impact whenever the additional information implied a downward update, that is, belief impact is signed in the direction of the rational update. Beliefs in Immediate serve as a benchmark that captures any difference in the effect of stories and statistics that is not related to memory.

The top panel of [Figure I](#) reveals that in line with our hypothesis, the decay in belief impact over time is substantially lower for stories than statistics. This is confirmed by [Table I](#), column (3). The difference-in-differences estimate of belief impact between Immediate and Delay is highly significant ($p < .01$).

We next consider point estimates of the belief impact in Immediate. The average belief impact in Immediate is larger for Statistic than for Story. On average, beliefs move by 21.76 percentage points (std. err. 0.58) for Statistic and by 18.19 percentage

points (std. err. 0.71) for Story.¹⁹ By contrast, for the Delay condition, the top panel of [Figure I](#) reveals that mean belief impact after one day is substantially more pronounced for Story than for Statistic. On average, belief impact was 5.93 percentage points (std. err. 0.73) in Statistic and 12.28 percentage points (std. err. 0.82) in Story. This divergence in belief impact in Delay is significantly different from zero ($p < .01$). [Online Appendix Figure A.10](#) underscores these patterns in the cumulative distribution functions of belief impact in Immediate and Delay, separately for stories and statistics.²⁰

2. *Recall.* It is conceivable that it may take some time for information to “sink in” and that the beliefs in Immediate are elicited before the information has been fully processed.²¹ In that case, using the immediate belief as a benchmark may not adequately capture the maximal belief update. We address these concerns using direct data on accurate recall of the provided information.

To study recall, we examine the fraction of respondents who correctly recall both the type and the direction of the information they were provided, as measured in the open-ended recall question. The bottom panel of [Figure I](#) shows that correct recall is significantly higher for stories than for statistics ($p < .01$). The average correct recall is 61.61% for stories and 28.70% for statistics. This suggests that information delivered in the form of stories is more easily retrieved than statistical information. Moreover, the open-ended data reveal several other patterns: (i) a large frac-

19. The immediate belief impact is close to the (average) Bayesian benchmark for both statistics (20.91 percentage points) and stories (18.71 percentage points). Note that for stories, we only consider the quantitative information contained in the review to compute the Bayesian benchmark, that is, we do not factor in the potential effect of the qualitative content provided. The experiment reported in [Section III.D](#) provides evidence from a setting where the Bayesian benchmark is well-defined also for stories.

20. The figure also highlights that there is substantial heterogeneity in perceived informativeness of the story treatment as measured in the immediate condition. This likely arises from differences in the way respondents interpret the qualitative information provided in the story.

21. In addition, there may be differences between stories and statistics in the extent to which participants rehearse the information they were provided with. Rehearsal cannot explain the findings of our mechanism experiments, presented in [Section IV](#), which hold constant the target information respondents are exposed to.

tion of respondents (44.91%) mention qualitative features from the story without specifically being prompted to do so; (ii) a very small fraction of respondents (1.32%) correctly recall and indicate the statistic they received; and (iii) only a negligible fraction (4.23%) mention the posterior belief they stated in the baseline wave. Note that this free-recall data provide a lower bound for whether people could recall each different feature of the information they were exposed to in wave 1. This is because they were not explicitly prompted to recall any specific feature.

3. *Heterogeneity by Extremity of Immediate Update.* Figure II illustrates the heterogeneity of delayed belief impact and correct recall by the extremity of the immediate update. The figure showcases little heterogeneity in correct recall by the extremity of the immediate belief update. For all levels of immediate updating, delayed belief impact and correct recall are higher for stories than for statistics.

RESULT 1. *There is a story-statistic gap in memory: over the course of one day, the effect of stories on beliefs decays less strongly than the effect of statistics. Stories have a stronger effect on beliefs than statistics in Delay, even though statistics have stronger immediate effects. Recall accuracy is substantially higher for stories than for statistics and does not depend on the strength of the immediate update.*

III.D. Robustness

1. *Experiment with Uninformative Qualitative Content.*

i. Design. A possible concern with our baseline design is that, as discussed, we cannot compute Bayesian benchmarks for the Story treatment. Respondents may interpret the qualitative content to be informative above and beyond the quantitative information. To deal with this concern, we conduct a robustness experiment that is identical to our baseline experiment except that it explicitly tells respondents that the qualitative content of the stories is uninformative, thus allowing for the computation of a Bayesian benchmark. We directly confirm that respondents understand this using an additional comprehension question. [Online Appendix A.A](#) provides more details.

ii. Sample. We recruited 1,000 respondents for the baseline survey; 912 respondents qualified for the follow-up survey. After

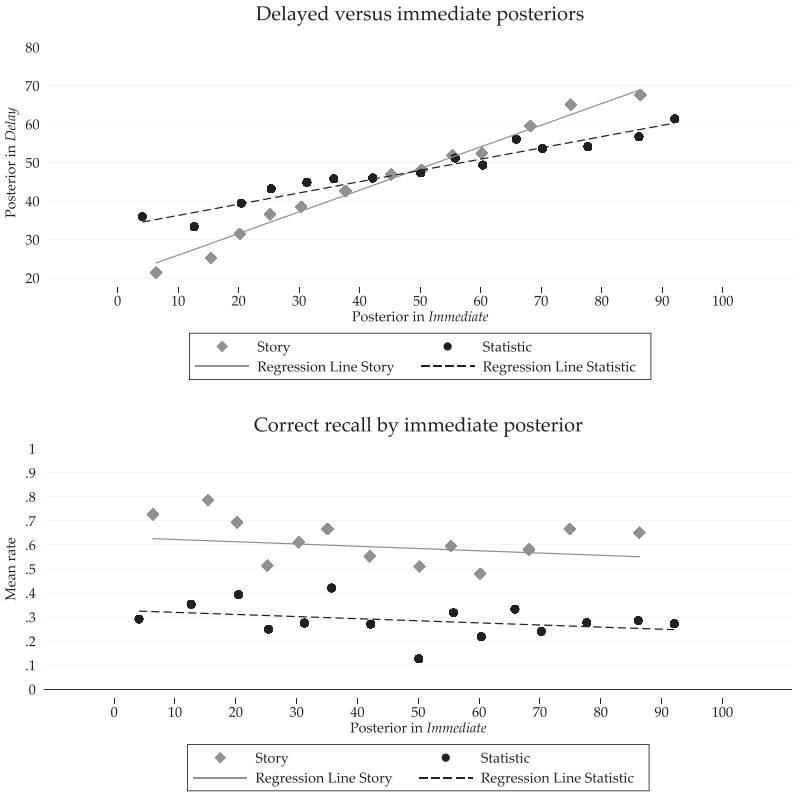


FIGURE II

Heterogeneity by Extremity of Immediate Update in the Baseline Experiment (933 Respondents)

The top panel displays binned scatterplots regressing beliefs in Delay (y-axis) on beliefs in Immediate, separately for conditions Story and Statistic. The bottom panel displays binned scatterplots regressing correct recall of the type and direction of information they received in the baseline survey in Delay (y-axis) on beliefs in Immediate, separately for conditions Story and Statistic. The black circles and line illustrate beliefs and recall for statistics, while the gray diamonds and line illustrate beliefs and recall for stories.

the prespecified sample restrictions, our final sample consists of 714 respondents, corresponding to a completion rate of 78%.

iii. Belief Movement. [Online Appendix](#) Figure A.1 confirms our baseline finding: The decay in belief impact over time is significantly lower for stories than statistics ($p < .01$), as illustrated by [Online Appendix](#) Table A.1, column (3).

We next consider point estimates of the belief impact in Immediate. Average belief impact in Immediate is larger for Statistic than for Story. On average, beliefs move by 21.91 percentage points (std. err. 0.49) for Statistic and by 19.21 percentage points (std. err. 0.51) for Story. The immediate belief impact is close to the average Bayesian benchmark for both statistics (22.07 percentage points) and stories (18.68 percentage points).²²

For the Delay condition, belief impact is 8.57 percentage points (std. err. 0.62) in Statistic and 10.52 percentage points (std. err. 0.66) in Story. This difference in belief impact in Delay is significantly different from zero ($p < .01$).²³ These findings underscore that respondents in the Story condition hold delayed beliefs that are significantly closer to the Bayesian benchmark.

iv. Incentivized Structured Recall. To complement the open-ended measure of recall from the baseline experiment, we use an incentivized structured recall task in this robustness experiment. We ask respondents to indicate whether they (i) received information about a single review, including some additional anecdotal details about the reviewer and their experience with the product; (ii) multiple reviews; (iii) no information; or (iv) don't know.²⁴ Unless respondents indicate that they did not receive any information about this product, we also ask them to indicate whether the information they received was positive or negative.²⁵

22. The difference between the two benchmarks might seem surprisingly small. Because a single review is either positive or negative, stories always have "extreme" realizations. Statistics, on the other hand, consist of more than one review and the fraction of positives can therefore also be moderate. For example, when exactly half the observed reviews are positive, the Bayesian belief movement equals zero. [Online Appendix F](#) shows that even for realizations with intermediate extremity, such as 75% positives, Bayesian belief movement is modest. Averaging over all statistics draws lumps together realizations of different extremity and leads to the relatively modest average belief impact seen in our sample. Put simply, all stories are extreme, while statistics are moderate on average.

23. This difference is smaller than in our baseline experiment possibly because respondents learn that the anecdotal details are not informative beyond the information contained in the quantitative reviews.

24. Respondents are told that if they choose "don't know," one of the other options will be randomly chosen to determine their payoff.

25. To keep the elicitation for stories and statistics as comparable as possible, we ask respondents who indicated that they had received multiple reviews whether the majority of reviews was positive or negative, while respondents indicating that they had received a single review were asked whether the single review was positive or negative. In this elicitation respondents can again select "don't know."

Respondents are told that if they correctly recall the information they received, they will obtain an additional bonus of \$5. To avoid hedging, either beliefs or recall were randomly selected for payment, and one question was randomly chosen to determine the bonus. The bottom panel of [Online Appendix Figure A.1](#) shows that correct recall is significantly higher for stories (69%) than for statistics (32%).

v. Willingness to Pay. To examine whether the information provided in our reviews would likely affect consumer behavior, we elicited a hypothetical willingness-to-pay measure after the belief elicitation. [Online Appendix A.2](#) shows that the Story and Statistic treatments both significantly increase willingness to pay compared to the no information treatment, in both the Immediate and Delay conditions. Given that the initial differences in willingness to pay are highly significant, these data do not lend themselves to studying memory mechanisms cleanly. Nonetheless, the additional measure suggests that the provided information is related to hypothetical willingness to pay.

2. Robustness to Other Design Features. In [Online Appendix A](#), we provide a series of robustness analyses. In [Online Appendix A.B](#), we show that manipulating the valence of story content only has very little effect on accurate recall. In [Online Appendix A.C](#) we examine the heterogeneity of our findings by positive versus negative information. [Online Appendix A.D](#) showcases the robustness of the story-statistic gap to different combinations of nontarget information. In [Online Appendix A.E](#), we examine how the size of the story-statistic gap depends on the number of products. Finally, [Online Appendix A.F](#) confirms the robustness of our findings to using different belief-elicitation formats.²⁶

III.E. Statistics with Qualitative Content

In line with our conceptualization of stories versus statistics, the story-statistic treatment variation from our baseline experiment changes both the number of reviews and varies the

26. In a previous version of this article, we conducted an additional experiment that varied the similarity between different scenario names (Restaurant A, Restaurant B, Restaurant C). This evidence shows that increasing prompt similarity decreases forgetting significantly. Because this design is not tightly connected to the conceptual framework anymore, this evidence is only described in the previous working paper version ([Graeber, Zimmermann, and Roth 2022](#)).

presence of additional qualitative content. In our model of selective memory, however, only the latter dimension, qualitative content, drives the story-statistic gap. To isolate the role of qualitative features, we conduct an additional experiment, in which information-free qualitative features are added to statistics.

1. *Design.* The incentives and basic setting closely follow the experiment described in [Section III.D](#). The only difference concerns the information respondents receive: for each product, participants receive either statistical information (condition *Statistic*), statistical information with an uninformative anecdote about one review (condition *Statistic with Qualitative Content*), or no further information. In the *Statistic with Qualitative Content* treatment, respondents are told that they receive additional details about one of the reviews. We use the same reviews as in the baseline experiment and always provide respondents with anecdotal details that are consistent with the direction—more positive or more negative reviews—implied by the statistic.

2. *Sample.* We recruited 1,000 respondents for the baseline survey; 906 respondents qualified for the follow-up survey. After the prespecified sample restrictions, our final sample consists of 673 respondents, corresponding to a completion rate of 74%.

3. *Results.* The top panel of [Figure III](#) as well as [Online Appendix Table A.9](#) confirm that the decay in belief impact over time is significantly lower for statistics with qualitative content than statistics without qualitative content ($p < .01$).²⁷ The bottom panel of [Figure III](#) shows that correct recall is significantly higher for statistics with qualitative content than for statistics alone. The average correct recall is 58.10% for statistics with qualitative content and 21.40% for statistics ($p < .01$). These findings confirm our model predictions and underscore the importance of qualitative content in driving the story-statistic gap in memory.

27. Immediate impact for statistics with qualitative content is larger than for statistics without qualitative content, even though all respondents passed an attention screen verifying that they understood that the qualitative information carries no additional information. This possibly arises from the qualitative stimuli enhancing the process of mental simulation ([Bordalo et al. forthcoming](#)).

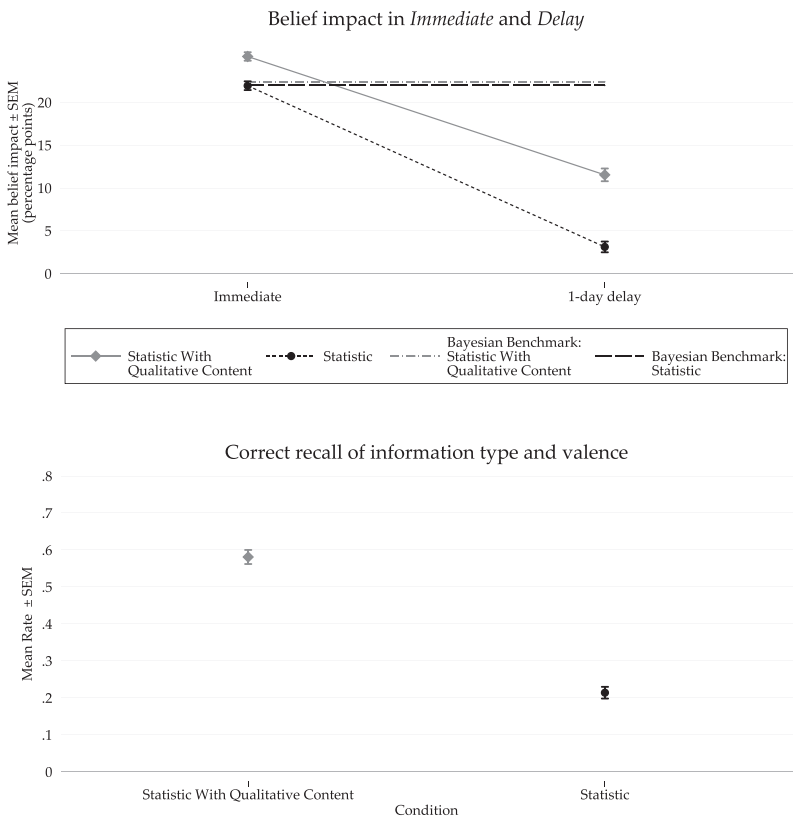


FIGURE III
Gap in Belief Impact and Recall for Statistics with and without Qualitative Content (673 Respondents)

The top panel displays belief impact in percentage points, separately for conditions Immediate and Delay. Belief impact is the signed distance between a stated belief and the prior (50%). Belief impact is signed in the direction of the rational update. The bottom panel displays the fraction of respondents correctly recalling the type and direction of information they received in the baseline survey. The gray markers illustrate belief impact and recall for statistics with qualitative content, while the black markers illustrate belief impact and recall for statistics. The gray horizontal line displays the average Bayesian benchmark for statistics with qualitative content (22.41 percentage points), while the black horizontal line illustrates the average Bayesian benchmark for statistics (21.98 percentage points). Whiskers indicate one standard error of the mean.

4. *Endogenous Qualitative Content.* In [Online Appendix A.G](#), we report the design and results of a related experiment. Instead of exogenously adding qualitative content to statistics, we ask respondents to imagine and describe a typical review in light of the statistical information provided. Our results indicate that prompting respondents to come up with a typical review when provided with a statistic increases delayed belief impact and improves recall accuracy, even though immediate updating remains unaffected by the prompt. Put differently, asking participants to add fictional qualitative features to a statistic on their own decreases the decay in belief impact. This further highlights the crucial role of qualitative content for the story-statistic gap.

III.F. Interpreting the Story-Statistic Gap

There are many differences between stories and statistics that our model abstracts from and may partly account for the story-statistic gap. In the following, we provide a brief overview and discussion.

1. *Engagement with Additional Information and Processing Time.* Differences in the processing time of stories and statistics, which may be indicative of the encoding strength, are a plausible mechanism underlying the story-statistic gap. We find that respondents spend somewhat more time processing stories (median of 42 seconds) than statistics (median of 32 seconds). [Online Appendix Table A.8](#) examines heterogeneity in belief impact and recall by the time spent processing the information. Correlationally, we find small and insignificant heterogeneity in differential belief impact based on initial processing time. Moreover, our mechanism experiments in [Section IV](#) hold the processing time of the target scenario constant, as they only vary similarity relationships between the prompt and the qualitative content of stories.

2. *Emotions and Vividness.* Research in psychology has established a connection between emotions and memory (e.g., [Kensinger and Schacter 2008](#)). Our evidence on the valence of story content ([Online Appendix A.B](#)) suggests that while stories with more consistent qualitative features are recalled at somewhat higher rates than stories with mixed and neutral qualitative story content, these differences are relatively small, especially

compared with the large differences in recall between stories and statistics. Moreover, while emotions plausibly play a role in driving the baseline story-statistic gap, the bulk of our mechanism evidence focuses on the features of cue-dependent memory, which allows us to hold emotions fixed.

3. *Outside Memories and Sample.* Respondents do not enter the experiment with a blank slate but bring in an outside database of memories. This existing database contains both stories and statistics to some extent, potentially affecting memory of different types of information. Observing information in the experiment (either in the form of a story or a statistic) might trigger the recall of such outside memories. These recollections from outside the experiment might, in turn, influence beliefs elicited in the experiment. In other words, outside memories might have a resonance effect, akin to information resonance in [Malmendier and Veldkamp \(2022\)](#). Our experimental instructions clarify that the only relevant information for the experimental tasks is the one provided within the experiment. Furthermore, it is important to note that such effects, if present, would already influence beliefs in the immediate elicitation. Hence, in order for resonance effects to explain the differences in the dynamic pattern of belief impact between stories and statistics, one would need to allow for such resonance effects to differ between Immediate and Delay. Alternatively, information resonance might shape how well the information is encoded and stored. Finally, our mechanism experiments, which hold the target information constant and operate by changing the similarity relationships, are immune to the information resonance mechanism.

IV. MECHANISMS

Guided by the predictions spelled out in [Section II](#), we proceed with an analysis of underlying mechanisms. First, we investigate the role of cue-target similarity in [Section IV.A](#). Then we examine interference in [Section IV.B](#).

IV.A. Cue-Target Similarity

Our model suggests that a key driving force behind the story-statistic gap is the similarity between the memory cue and the target information. The qualitative content of stories is often

related in meaning to the memory cue that originates from the prompt. Therefore, stories more easily come to mind than statistics, which, due to their abstract nature, tend to be unrelated to memory cues. To examine the role of cue-target similarity, we conduct experiments that manipulate the degree of similarity between stories and memory cues arising from the experimental prompt.

1. *Design.* The incentives and basic setting closely follow the experiment described in [Section III.D](#). Unlike in the baseline experiments where information types are randomized across scenarios, here we always provide a story in the restaurant scenario. Our key treatment varies between subjects the similarity between the story for that scenario and the prompt, that is, the name of the scenario, which is shown together with the belief elicitation question (in both the Immediate and Delay conditions). Specifically, for the restaurant scenario, each respondent receives either a positive or a negative story. The story describes a reviewer's experience in an Italian restaurant, where the experience makes explicit reference to the Italian cuisine of the restaurant. The scenario identifier, that is, the label for the restaurant, appears at the top of the belief elicitation page and does not appear in the story itself.²⁸ In the other two scenarios, respondents receive a statistic and no additional information. We then vary across participants how similar the prompt in the restaurant scenario is to the content of the story. We have three treatment conditions. In the High Similarity condition, the name of the restaurant is The Italian restaurant "Napoli". In Low Similarity 1, the name of the restaurant is An eatery, while in Low Similarity 2 the name of the restaurant is Mr. Jones. The high-similarity prompt mentions the specific type of Italian cuisine, while the low-similarity prompts have no direct association with Italian food. Low Similarity 1 is a generic term for a dining establishment, while Low Similarity 2 is a generic name that does not even reveal that the venue of interest is a restaurant.²⁹ Our

28. [Online Appendix C.B](#) reproduces the stories.

29. We validate these intuitions by computing the cosine-similarity between the prompts and the story. Using a tf-idf vectorization with the three prompts and the story as corpus, we obtain a similarity with the story of 0.31 for The Italian restaurant "Napoli," against 0.00 for both An eatery and Mr. Jones. Using a vectorization based on OpenAI's state-of-the-art embedding model, text-embedding-

design thus varies the experimental prompt, which underlies the memory cue in our model.

2. *Sample.* We recruited 1,000 respondents for the baseline survey; 912 respondents qualified for the follow-up survey. After the prespecified sample restrictions, our final sample consists of 627 respondents, corresponding to a completion rate of 69%.

3. *Result.* [Figure IV](#) and [Online Appendix Table A.10](#) present the results from this experiment. The upper panel shows results on immediate and delayed belief movement. The figure illustrates that the decay in belief impact over time is significantly lower for High Similarity than for Low Similarity 1 ($p < .01$) and Low Similarity 2 ($p < .01$). The comparison of belief decay is straightforward as beliefs in Immediate do not differ significantly.

The lower panel of [Figure IV](#) confirms these patterns in the recall data. While accurate recall in High Similarity is at 80%, it is at only 70% and 38% in Low Similarity 1 and Low Similarity 2, respectively. These effect sizes are large in magnitude and highly statistically significant ($p < .01$). Taken together, these results underscore the quantitative importance of the cue-target similarity mechanism.

Reassuringly, [Online Appendix Figure A.11](#) shows no significant differences in the belief movement and recall for statistics across the three treatments. This suggests that, as intended, our manipulation only affected recall of the restaurant scenario. Moreover, this pattern allows us to cleanly cast our findings in terms of the story-statistic gap. Increasing cue-story similarity significantly increases the story-statistic gap in memory.

RESULT 2. *Increases in cue-target similarity significantly increase delayed belief impact and recall accuracy.*

IV.B. Similarity of Cue to Nontarget Information

While the qualitative content of stories tends to give them an edge over statistics due to its natural similarity to the cue, our model also clarifies that qualitative content of nontarget information can inhibit the recall of target information due to

3-large, we obtain a similarity with the story of 0.52 for The Italian restaurant “Napoli,” compared with 0.28 for An eatery and 0.10 for Mr. Jones.

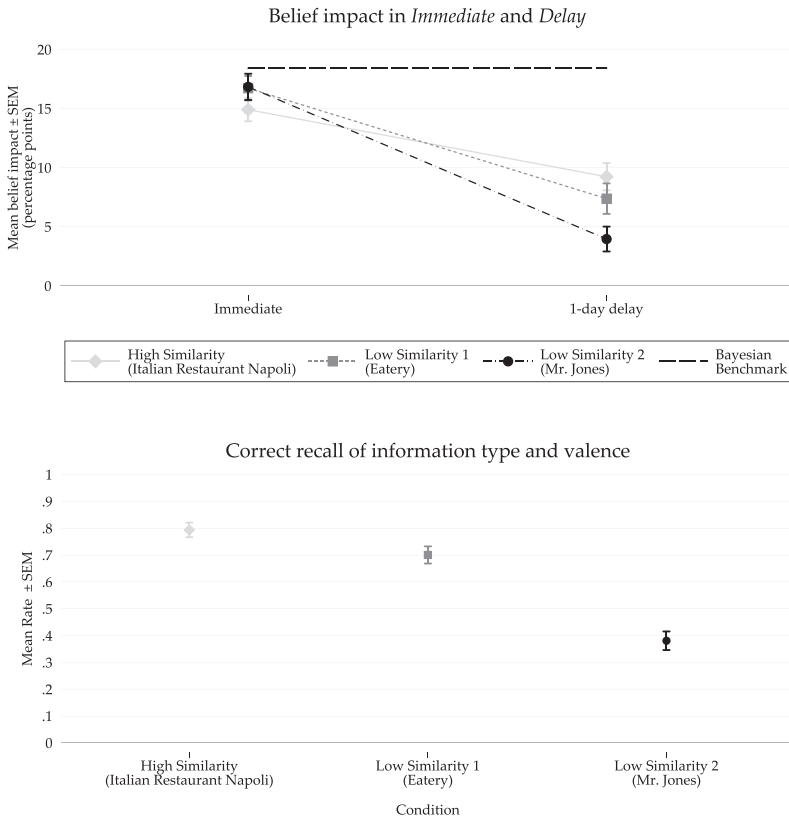


FIGURE IV

Belief Impact and Recall in Mechanism Experiment 1 (670 respondents)

The top panel displays belief impact in percentage points, separately for conditions Immediate and Delay. Belief impact is the signed distance between a stated belief and the prior (50%). Belief impact is signed in the direction of the rational update. The bottom panel displays the fraction of respondents correctly recalling the type and direction of information they received in the baseline survey. The light gray markers illustrate belief impact and recall for High Similarity, the dark gray illustrate belief impact and recall for Low Similarity 1, while the black markers show belief impact and recall for Low Similarity 2. The horizontal black dashed line illustrates the Bayesian benchmark (18.45 percentage points). Whiskers indicate one standard error of the mean.

interference. Specifically, a direct prediction of our model is that a higher similarity between the memory cue and nontarget stories creates memory interference in the recall of target information and reduces accurate recall after some delay. To test this predic-

tion, this experiment directly manipulates the similarity between the experimental prompt and a set of nontarget stories.

1. *Design.* The incentives and basic setting are identical to the mechanism experiment presented in the previous section. We have two treatment conditions that vary, between participants, the similarity of the prompt to nontarget information. All participants learn about three scenarios: a food truck, an amusement park, and a sports stadium. Unlike in our main experiment, respondents receive a story in each of the three scenarios. In Low Interference, the three stories are distinct and specific to each prompt. Our target story concerns a food truck and describes the quality of a hot dog. Our nontarget stories concern a sports stadium and an amusement park, which describe features of the stadium and the amusement park, respectively. In the High Interference condition, we keep the target story about the food truck identical to Low Interference but increase the similarity of the two nontarget stories to the food truck prompt. In High Interference, the two other prompts are still about an amusement park and a sports stadium, but now their corresponding stories describe hot dogs consumed at these locations.³⁰ Our design thus varies the experimental prompt, which underlies the memory cue in our model.

Thus, our experiment fixes the target story and only manipulates the similarity between the two nontarget stories and the food truck prompt.³¹ All other design aspects are identical across conditions. [Online Appendix C.C](#) reproduces the stories that we used.

2. *Sample.* We recruited 1,000 respondents, of which 670 qualified for the follow-up.³² After the prespecified sample restric-

30. We compute the cosine-similarity between the prompts and the nontarget stories across the two treatments with the help of a large-language model. Using a vectorization based on OpenAI's state-of-the-art embedding model, text-embedding-3-large, yields an average similarity of the prompt with the stories in High Interference of 0.25 compared with 0.11 in Low Interference. This validates our intuitions about the similarity relationships.

31. The target story concerns a positive review, whereas the nontarget stories both feature a negative review.

32. The somewhat larger fraction of respondents not qualifying for the follow-up study can be explained by them failing our prespecified inclusion criterion of updating in the right direction in Immediate in all three scenarios.

tions, we have a sample size of 505, corresponding to a completion rate of 75%.³³

3. *Results.* The top panel of Figure V shows data on the belief impact of the target story in Immediate and Delay, separately for High Interference and Low Interference. While there is no difference in belief impact in Immediate, the slope in belief impact is steeper in High Interference compared with Low Interference, in line with the model prediction. Delayed belief impact is significantly lower in High Interference than in Low Interference.

While average delayed belief impact in High Interference is 5.91 percentage points (std. err. 1.05), it is 9.86 percentage points (std. err. 1.12) in Low Interference. Online Appendix Table A.10 confirms this visual pattern and shows that the difference-in-differences in belief impact (difference in slopes) is statistically significant ($p < .01$).³⁴

The bottom panel illustrates similar patterns for recall: among respondents in Low Interference, 48.36 percentage points (std. err. 3.43) correctly recall the information, compared with only 31.16 percentage points (std. err. 2.72) in High Interference. This difference of 17.2 percentage points is statistically significant ($p < .01$). This effect size is moderate in size and corresponds to 0.35 of a standard deviation.³⁵

RESULT 3. *Increases in similarity of the prompt to nontarget information significantly decreases delayed belief impact and recall accuracy of the target information.*

4. *Implications.* This finding has several implications. First, it provides strong evidence for the power of similarity relationships in determining the decay of belief impact and recall ac-

33. The completion rate to the follow-up survey does not differ significantly across treatment groups ($p = .62$).

34. In a previous version of the article, we confirm the robustness of these results on interference using another experiment that featured a different set of prompts and stories but overall employed a similar design (Graeber, Zimmermann, and Roth 2022).

35. Forgetting in Low Interference of this mechanism experiment is higher than in our baseline experiment for potentially two reasons. First, the pieces of additional information (three stories) may be more similar to one another than the information provided in the baseline experiment. Second, respondents in this mechanism experiment receive three pieces of information instead of only two pieces of information in the baseline experiment.

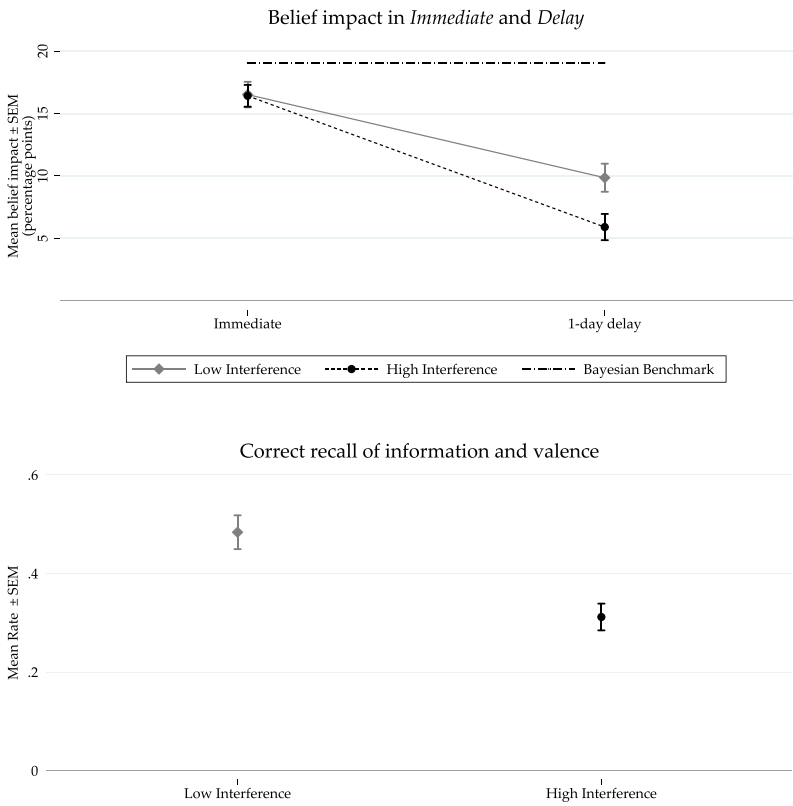


FIGURE V
Belief Impact and Recall in Mechanism Experiment 2 (505 respondents)

The top panel displays belief impact in percentage points, separately for conditions Immediate and Delay. Belief impact is the signed distance between a stated belief and the prior (50%). Belief impact is signed in the direction of the rational update. The bottom panel displays the fraction of respondents correctly recalling the type and direction of information they received in the baseline survey. The gray markers illustrate belief impact and recall for Low Interference, while the black markers illustrate belief impact and recall for High Interference. Whiskers indicate one standard error of the mean.

curacy. Second, it delineates the limits of the stickiness of stories in memory. If the memory database contains many stories that are similar to a cue, retrieval of a target story gets crowded out and it becomes less likely that this story comes to mind. Hence, stories as a communication device lose their edge in environments where similar stories circulate.

V. DECOMPOSING THE STORY-STATISTIC GAP

The evidence presented so far leaves some fundamental questions on the processes underlying selective memory unaddressed. One distinction with far-reaching consequences for both theoretical and empirical work is whether distortions induced by memory result from (i) successfully retrieving memories that are subject to partial information loss or (ii) complete retrieval failures. In the following, we provide a decomposition with the purpose of quantifying the importance of these two margins of memory in driving the story-statistic gap.³⁶

V.A. *Three Variants of Memory Distortions*

To examine these ideas, note that the basic memory retrieval process can have three different possible outcomes: (i) retrieval failure, (ii) successful retrieval without information loss, and (iii) successful retrieval of a memory trace that is subject to information loss.³⁷ Each outcome is associated with a different signature in the decay of belief impact. First, the DM may not retrieve any target memory and beliefs therefore revert to the prior (class FullDecay). Second, the DM may correctly recall the full wealth of information contained in a story or statistic (class ZeroDecay). In that case, the DM would state their past posterior belief and we would observe no decay in beliefs over time. Third, the DM may successfully recall the target memory trace, but the retrieved information is subject to information loss (class IntermediateDecay). The corresponding signature in beliefs would be a partial reversion to the prior.

V.B. *Empirical Approach*

The combination of recall data and the evolution of beliefs allow us to shed some light on the relative importance of these margins of memory distortions. We proceed as follows. We use recall data to identify retrieval failures and test whether this class is, in fact, associated with FullDecay in the corresponding stated beliefs. We turn to the remaining data, which by construction only

36. The analyses in this section are exploratory in nature and were not pre-registered.

37. The latter could be captured by a model that also incorporates noise in the belief formation or retrieval process, along the lines of [Enke and Graeber \(2023\)](#); [Ba, Bohren, and Imas \(2023\)](#).

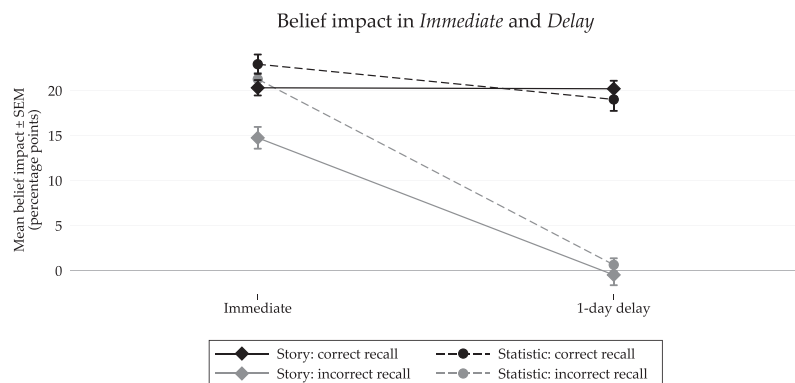


FIGURE VI

The Decay of Belief Impact by Recall Accuracy in the Baseline Experiment (933 Respondents)

The figure displays belief impact in percentage points, separately for conditions Immediate and Delay. Belief impact is the signed distance between a stated belief and the prior (50%). Belief impact is signed in the direction of the rational update. The black diamond markers illustrate belief impact for stories with correct recall, while the gray diamond markers illustrate belief impact for stories with incorrect recall. The black circle markers illustrate belief impact for statistics with correct recall, while the gray circle markers illustrate belief impact for statistics with incorrect recall. Whiskers indicate one standard error of the mean.

include observations where people correctly remember at least some of the information—information type and direction. We focus on the corresponding belief data to assess the relative shares of ZeroDecay and IntermediateDecay. The following analysis focuses on our baseline experiment reported in [Section III](#).

V.C. Results

First, recall that the bottom panel of [Figure I](#) identifies the fraction of beliefs associated with retrieval failure, which is defined as incorrect recall of the type and/or direction of the additional information. Following this metric, 38% of observations in Story, but 71% in Statistic fail to retrieve relevant information about the target trace. According to the model, these observations should be associated with beliefs that fully revert to the prior of 50%, implying a belief impact of zero (class FullDecay). [Figure VI](#) displays the story-statistic gap in belief impact separately for the sample of observations with correct and incorrect recall (following the definition of the bottom panel of [Figure I](#)). The average belief

impact for observations classified as FullDecay indeed reverts to close to zero in Delay, as predicted by the model.

Second, turning to observations with correct recall of type and direction, we can identify the class of ZeroDecay as those that state identical beliefs in Immediate and Delay. This constitutes 37.48% (46.88% of correct recall observations) of all observations in Story and 27.79% (56.05% of correct recall observations) of all observations in Statistic. Note that these figures likely identify a lower bound, because they do not take into account potential measurement error in beliefs. If people in ZeroDecay answer the belief questions with some added random noise, there would be no average belief decay, yet many would state beliefs that differ between the two periods.³⁸

Finally, we turn to the remaining class, IntermediateDecay, with correct recall of type and direction, but at the same time featuring beliefs with some intensive-margin information loss by virtue of being part of neither ZeroDecay nor FullDecay. Above we already classified a substantial lower bound for the class ZeroDecay. Figure VI displays average belief decay among observations with correct recall of type and direction. Strikingly, it reveals that there is zero average belief decay in the Story condition and a quantitatively minor, only marginally significant decay in the Statistic condition. Put differently, conditional on correct recall, we see close to no evidence for belief decay, suggesting a central role of retrieval failures.

V.D. Interpretation

Taken together, this exercise provides a clear conclusion. In our experiments, patterns of selective memory are driven by a failure to retrieve any relevant memory for a given scenario, rather than successful recall with partial information loss.

V.E. Within-Person Variation

To complement this evidence, we more systematically examine the within-person movements in beliefs between Immediate and Delay. To illustrate the connection between second-period beliefs, first-period beliefs, and the ignorance prior, we calculate an individual belief decay parameter, λ , for each participant using

38. We can instead apply a more lenient benchmark than precisely zero decay, but as will be clear below, this will, if anything, only strengthen our conclusion.

the following specification:

$$\text{Belief}_{i2} = \lambda \times \text{IgnorancePrior} + (1 - \lambda) \times \text{Belief}_{i1},$$

where Belief_{i2} is the second-period belief and Belief_{i1} is the first-period belief. λ takes value one for respondents returning to the ignorance prior and zero for respondents with perfect recall. [Online Appendix](#) Figure A.12 displays the belief-decay estimates for our main experiment. Panel A displays results pooling data from the Story and Statistic condition, while Panel B displays results for these conditions separately. Panel A highlights that a large fraction of the data is at $\lambda = 1$ and $\lambda = 0$, that is, full forgetting or full recall. This highlights an all-or-nothing nature of recall in this setting. Panel B shows that there is more mass at $\lambda = 0$ and less mass at $\lambda = 1$ in the Story condition compared with the Statistic condition. In other words, respondents in the Statistic condition are much more likely to state the ignorance prior compared to people in the Story condition. Future work should examine the robustness of the all-or-nothing nature of recall in this setting. For example, it is worth exploring whether “partial information loss” becomes more important when there is a longer delay between the receipt and recall of the information.

VI. DISCUSSION AND CONCLUSION

This article documents a story-statistic gap in memory. As time passes, the effect of information on beliefs generally decays, but this decay is much less pronounced for stories than for statistics. Using recall data, we show that stories are more accurately retrieved from memory than statistics. We causally show that this pattern is driven by the presence of qualitative content in stories. Guided by a simple model of cue-dependent memory, we experimentally demonstrate the explanatory power of two key forces of memory: cue-target similarity and interference. Our memory decomposition provides striking evidence that retrieval failures appear to be the key driver of the story-statistic gap, rather than partial information loss in retrieved memories.

1. *Stories in the Mass Media.* Our findings have implications for understanding several real-world phenomena. Mass media not only provides facts and statistics but also relies on anecdotes about individual cases, which provide detailed qualitative information. Consider allegations about election fraud in the con-

text of the 2020 U.S. presidential election, where some outlets reported stories about individual cases of election fraud, even though these were rare exceptions. Likewise, consider news reporting about welfare fraud, where anecdotes about individual cases are abundant in the news media but stand in stark contrast to official statistics on fraud incidence. For example, Ronald Reagan, beginning with his 1976 campaign, told extreme stories about “welfare queens”:

She has 80 names, 30 addresses, 12 Social Security cards and is collecting veterans' benefits on four non-existing deceased husbands. And she's collecting Social Security on her cards. She's got Medicaid, getting food stamps, and she is collecting welfare under each of her names. Her tax-free cash income alone is over \$150,000.

Similar patterns emerge in mass media coverage of immigration. While statistics about low crime rates among immigrants are widely reported by news outlets, extreme stories about immigrants committing serious crimes also regularly hit the headlines. Our results indicate that stories disseminated in this way can have powerful effects on beliefs because they may come to mind more easily than more representative statistical information.

2. *Policy Communication.* Our results also have implications for the communication of statistical information. If policy makers, marketers, or leaders aim to convey statistical information effectively, they may wish to complement it with anecdotes to ensure that the information sticks with the audience. For instance, statistical information about economic quantities could be coupled with anecdotal information that is consistent and inherently reminiscent of the embedded statistical information. Moreover, our results suggest that persuaders should factor in the time structure when picking their mode of persuasion: if messaging occurs close in time to the audience's anticipated action, statistics and quantitative facts can be more powerful than stories; as soon as a delay is involved, stories trump statistics.

3. *Avenues for Future Research.* First, it would be desirable to gain a better understanding of the evolution of memory patterns and the story-statistic gap as time delays increase. Second, to shed light on the external validity of our findings, it will be important to assess whether our results are specific to the recall of statistical information or whether they instead extend to the

recall of simple facts devoid of any context. Third, it would be interesting to understand how memory mechanisms affect the virality of different types of information. Finally, future work might examine whether there is also a gap in the evolution of beliefs between personal and nonpersonal experiences that is analogous to the story-statistic gap.

SUPPLEMENTARY MATERIAL

An Online Appendix for this article can be found at *The Quarterly Journal of Economics* online.

DATA AVAILABILITY

The data underlying this article are available in the Harvard Dataverse, <https://doi.org/10.7910/DVN/AOY4D6> (Graeber, Roth, and Zimmermann 2024).

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